

14.2.5.7 Fourier Transforms and Convolution

It is interesting to note that convolution in the time domain corresponds to multiplication in the frequency domain and vice versa. Given a system whose impulse response is $h(t)$, we know that we can find the output of the system $y(t)$ in response to an input $x(t)$ as follows:

$$y(t) = x(t) * h(t).$$

In the frequency domain, given the Fourier transforms of the impulse response $H(\omega)$ and the input $X(\omega)$, we can find the Fourier transform of the output as follows:

$$Y(\omega) = X(\omega)H(\omega).$$

This result is pretty incredible, and it's also very handy. Sometimes it is more convenient to perform a convolution on the time axis using a system's impulse response $h(t)$, while at other times it's more convenient to perform a multiplication in the frequency domain using the system's *frequency response* $H(\omega)$.

As it turns out, LTI systems exhibit a property called *duality*, which says that you can reverse the roles of time and frequency and virtually the same mathematical rules continue to apply. So, for example, we can understand how signal *modulation* (the multiplication of one signal by another) works in the time domain by looking at what happens when we convolve the Fourier transforms of the two signals on the frequency axis. Having two ways of tackling a problem is always better than one!

14.2.5.8 Filtering

The Fourier transform allows us to visualize the set of frequencies that make up virtually any audio signal. A *filter* is an LTI system that attenuates a selected range of input frequencies while leaving all other frequencies unaltered. A *low-pass* filter retains low frequencies while attenuating high frequencies. A *high-pass* filter does the opposite, retaining high frequencies and attenuating lower frequencies. A *band-pass* filter attenuates both low and high frequencies but retains frequencies within a limited *passband*. A *notch* filter does the opposite, retaining low and high frequencies but attenuating frequencies within a limited *stopband*.

Filters are used in a stereo system's *equalizer*, by attenuating or boosting specific frequencies based on user inputs. Filters can also be used to attenuate noise, if the spectra of the noise signal and the desirable signal occupy different regions of the frequency axis. For example, if a high-frequency noise signal is

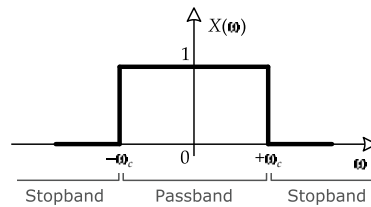


Figure 14.21. The frequency response $H(\omega)$ for an ideal filter has a value of one in the passband and zero in the stopband.

adversely affecting a lower-frequency voice or music signal, a low-pass filter could be used to eliminate the noise.

The frequency response $H(\omega)$ of an ideal filter looks like a rectangular box, with a value of one in the passband and zero in the stopband. When we multiply this by the Fourier transform of our input signal $X(\omega)$, the output $Y(\omega) = X(\omega)H(\omega)$ will have its passband frequencies preserved exactly, and its stopband frequencies all set to zero. The frequency response for an ideal filter is shown in Figure 14.21.

Of course, an ideal filter that completely passes certain frequencies and completely suppresses others may not be desirable. The frequency responses of most real-world filters have a gradual fall-off between the passband and stopband. This aids filtering in situations where there is no single, clear-cut line between the desirable frequencies and the unwanted frequencies. The frequency response of a low-pass filter with a gradual fall-off is shown in Figure 14.22.

The *equalizer* (EQ) found on most high-fidelity audio equipment permits the user to adjust the amount of bass, mid-range and treble that is output. An EQ is really just a collection of filters tuned to different frequency ranges and applied in series to an audio signal.

Filtering theory is an immense field of study, so we can't possibly do it justice here. For a great deal more information, see [41, Chapter 6].

14.3 The Technology of Sound

Before we can fully understand the software that comprises a game's audio engine, we need a firm grasp of audio hardware and technology and of the terminology used by industry professionals to describe it.

14.3.1 Analog Audio Technology

The earliest audio hardware was based on analog electronics. This was the easiest way to record, manipulate and play back audio compression waves, because sound is itself an analog physical phenomenon. In this section, we'll briefly explore some key analog audio technologies.

14.3.1.1 Microphones

A microphone (also known as a “mic” or “mike”) is a transducer that converts an audio compression wave into an electronic signal. Microphones make use of various technologies in order to convert the mechanical pressure variations of a sound wave into an equivalent signal based on variations in electric voltage. A *dynamic microphone* uses electromagnetic induction, while a *condenser microphone* utilizes changes in capacitance. Other types of mics use piezoelectric generation or light modulation to produce a voltage signal.

Different microphones have different sensitivity patterns, known as *polar patterns*. These patterns describe how sensitive the mic is to sound at various angles about its central axis. An omnidirectional mic is equally sensitive in all directions. A bidirectional mic has two sensitivity “lobes” in the shape of a figure eight. A cardioid mic has essentially a unidirectional sensitivity profile, so named because of its somewhat heart-shaped polar pattern. Some common

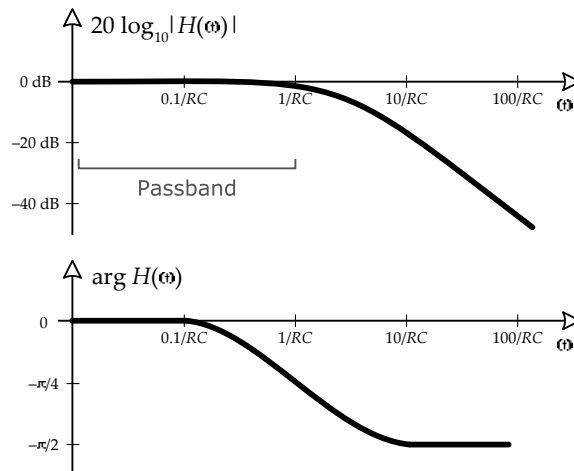


Figure 14.22. The frequency response $H(\omega)$ for an RC (resistor-capacitor) low-pass filter with a gradual fall-off. Both the horizontal and vertical axes of both plots are drawn using a logarithmic scale.

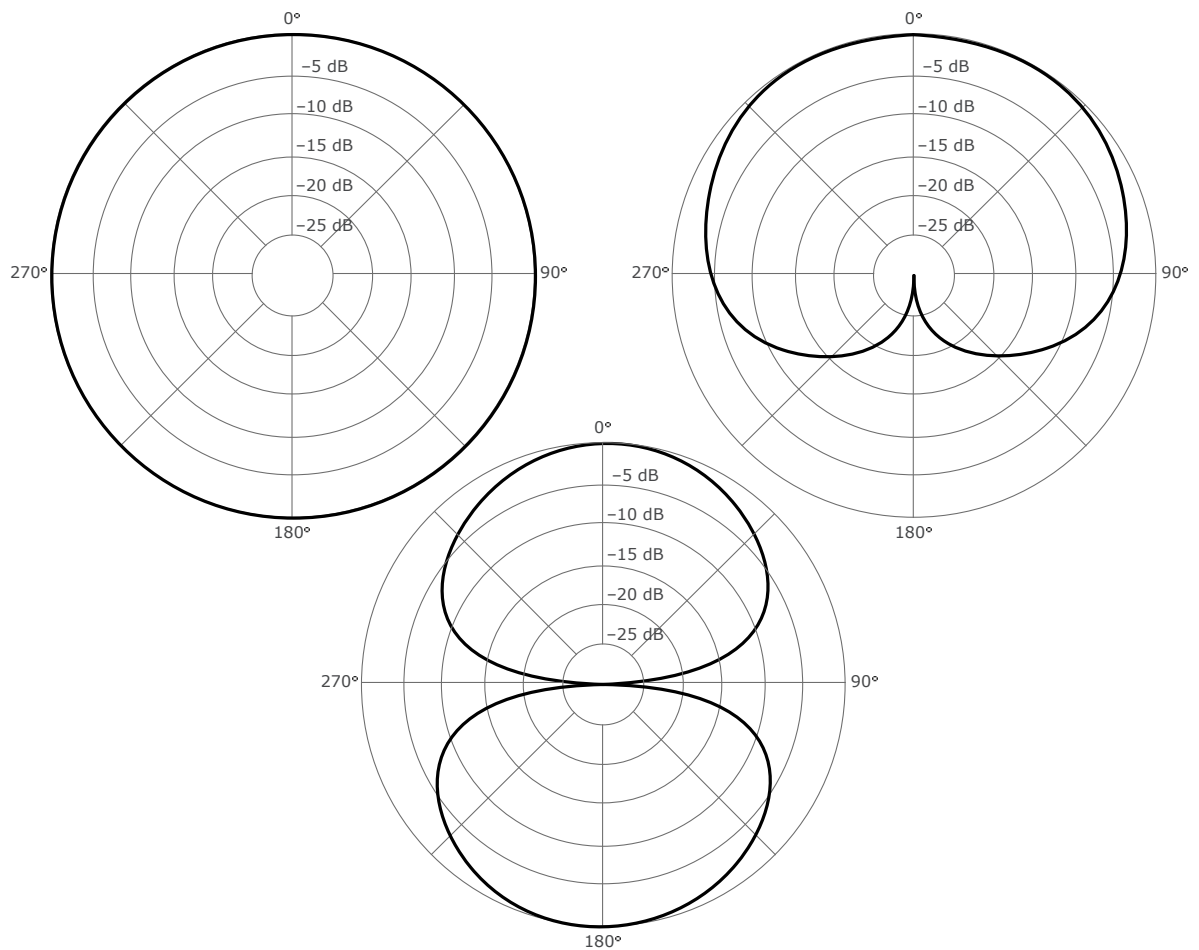


Figure 14.23. Three typical microphone polar patterns, clockwise from upper left: omnidirectional, cardioid and bidirectional.

microphone polar patterns are illustrated in Figure 14.23.

14.3.1.2 Speakers

A *speaker* is basically a microphone operated in reverse—it is a transducer that converts a varying input voltage signal into vibrations in a membrane, which in turn gives rise to air pressure variations that result in a sound pressure wave.

14.3.1.3 Speaker Layouts: Stereo

Sound systems usually support multiple speaker output channels. A stereo device such as an iPod, the sound system in your car or your grandpa's portable "boom box" supports at least two speakers for the left and right stereo channels. Some high-fidelity stereo systems also boast two additional "tweeters"—tiny speakers that are capable of reproducing the highest-frequency sounds within the left and right channels. This allows the two main speakers to be larger, and therefore better at covering the bass. Some stereo systems also support a subwoofer or LFE (low-frequency effects) speaker. Such systems are sometimes called 2.1 systems—two for the left and right, and "dot one" for the LFE speaker.

Headphones versus Speakers

It's important to distinguish between stereo speakers in an open room and stereo headphones. Stereo speakers in a room will typically be positioned in front of the listener and offset to either side. This means that the sound waves coming from the *left* speaker are actually received by the *right* ear as well, and vice versa. The waves from the more-distant speaker will arrive at the ear with a slight time delay (phase shift) and a slight attenuation. The phase-shifted sound waves from the more-distant speaker will tend to *interfere* with those coming from the closer speaker. The sound system should take this interference into account in order to produce the highest quality sound.

Headphones, on the other hand, come in direct contact with the ears, so the left and right channels are perfectly isolated and do not interfere with one another. Also, because headphones deliver sound almost directly to the ear canal, the head-related transfer effects (HRTF) of the shape of the ears themselves do not come into play (see Section 14.1.4), meaning that somewhat less spatial information is received by the listener.

14.3.1.4 Speaker Layouts: Surround Sound

Home theater *surround sound* systems typically come in two flavors: 5.1 and 7.1. As you undoubtedly guessed, these numbers refer to the five or seven "main" speakers, plus the one subwoofer. The goal of a surround sound system is to immerse the listener in a realistic soundscape, by providing *positional* information as well as high-fidelity sound reproduction (see Section 14.1.4). The main speaker channels in a 5.1 system are: center, front left, front right, rear left and rear right. A 7.1 system adds two additional speakers, surround left and surround right, which are intended to be placed directly to either side of the listener. Dolby Digital AC-3 and DTS are two popular surround sound

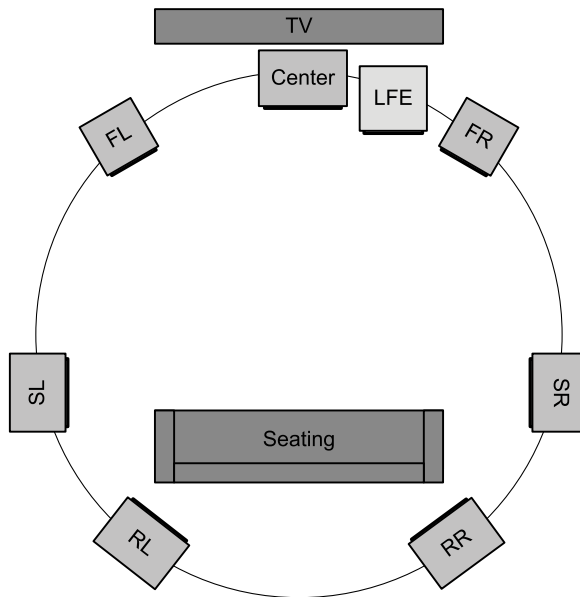


Figure 14.24. Speaker arrangement for a 7.1 surround sound home theater system.

technologies. The speaker layout of a typical 7.1 home theater is shown in Figure 14.24.

Dolby Surround, Dolby Pro Logic and Dolby Pro Logic II are technologies for expanding a stereo source signal into 5.1 surround sound. A stereo signal lacks the positional information necessary to drive the 5.1 speaker configuration directly. But using these Dolby technologies, an approximation of the missing positional information can be generated heuristically using various cues found within the original stereo source signal.

14.3.1.5 Analog Signal Levels

Audio voltage signals may be transmitted at various voltage levels. A microphone usually produces a low-amplitude voltage signal—these are called *mic-level* signals. For connections between components, higher-voltage *line-level* signals are used. There's a big difference between professional audio equipment and consumer electronics when it comes to line-level voltages. Professional gear is usually designed to work with line levels ranging from 2.191 V (volts) peak-to-peak for a nominal signal up to a maximum voltage of 3.472 V peak-to-peak. The peak-to-peak voltage of a “line level” signal on consumer equipment varies quite a bit, but most consumer devices output up to 1.0 V peak-to-peak, and have inputs that can handle up to 2.0 V signals. It's impor-

tant to match the levels of input and output signals when connecting audio equipment. Passing a voltage that is too high for the device to handle will cause clipping of the signal. And passing a voltage that is too low will result in audio that sounds quieter than it should.

14.3.1.6 Amplifiers

The small voltages produced by a microphone must be *amplified* in order to drive speakers with enough force to produce audible sound waves. An *amplifier* is an analog electronic circuit that produces at its output a nearly exact replica of its input signal, but with the *amplitude* of the signal increased significantly. An amp essentially increases the *power* content of a signal. It does this by drawing from some kind of power source, and driving the increased voltage produced by this power source in such a way as to mimic the behavior of the input signal over time. In other words, an amp *modulates* the output of its power source to match its much lower-voltage input signal.

The core technology behind an amplifier is the *transistor*—that well-known and utterly ingenious device that sits at the heart of many modern electronic devices, including its crowning achievement—the computer. A transistor makes use of a semiconducting material in order to link the voltages between two otherwise isolated, independent circuits. As such, a low-voltage signal can be used to drive a higher-voltage circuit. This is exactly what is required of an amplifier. We won't get into the details of how transistors and amplifiers work under the hood here. But if you're curious, you can whet your whistle with this great YouTube video on how the very first transistor worked: <https://www.youtube.com/watch?v=RdYHljZi7ys>. And you can read more about amplifier circuits here: <http://en.wikipedia.org/wiki/Amplifier>.

The *gain* A of an amplification system is defined as the ratio of output power P_{out} to input power P_{in} . Like sound pressure level, gain is typically measured in decibels:

$$A = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right) \text{ dB.}$$

14.3.1.7 Volume/Gain Controls

A *volume control* is basically an inverse amplifier, also known as an *attenuator*. Rather than increasing the amplitude of an electrical signal, it *decreases* the amplitude, while keeping all other aspects of the waveform intact. In a home theater system, the D/A converter produces a voltage signal with a very small amplitude. The power amp boosts this signal up to the maximum “safe” output power, beyond which the sound produced by your speakers would begin to clip and distort (or even damage your hardware). The volume control then

attenuates this maximum output power to produce sound at the desired listening volume.

A volume control is much simpler to make than an amplifier. One can be constructed by introducing a variable resistor, also known as a *potentiometer*, into the circuit somewhere between the amplifier's output and the speakers. When the resistance is at its minimum (at or very close to zero), the amplitude of the input signal isn't changed, and a sound of maximum volume is produced. When the resistance is at its maximum setting, the input signal's amplitude is maximally attenuated, and a sound of minimum volume is produced.

If your stereo system at home reports the volume in decibels, you've probably noticed that the values are always negative. This is because the volume control is attenuating the output of the power amp. The volume meter is still measured like a gain, but the "input" power is the maximum power of the amp, and the "output" power is the volume selected by the user:

$$A = 10 \log_{10} \left(\frac{P_{\text{volume}}}{P_{\text{max}}} \right) \text{ dB},$$

which will be negative as long as $P_{\text{volume}} < P_{\text{max}}$.

14.3.1.8 Analog Wiring and Connectors

An analog monophonic audio voltage signal can be carried by a pair of wires; a stereo signal requires three wires (two channels plus a common ground). The wiring can be internal to a device, in which case it is usually called a *bus*. Wiring can also be external, for use in connecting different devices to one another.

External wiring is typically connected to audio hardware either via a direct "clip" or screw-post connector, of the kind found on high-end speakers, or via various standardized connectors. Examples include RCA jacks, large TRS (tip/ring/sleeve) jacks (the kind used by telephone operators in the early 1900s), TRS mini-jacks (found on your iPod, mobile phone and most PC sound cards), keyed jacks (found most often on high-quality microphones and power amps), and the list goes on.

Audio wiring is available in a wide range of quality levels. Thicker-gauge wiring offers less resistance and therefore can transmit signals over farther distances without unacceptable levels of attenuation. Optional shielding can help reduce noise. And of course the choice of which metal to use in the construction of the wires and connectors can make a difference in the quality of the wiring as well.

14.3.2 Digital Audio Technology

The introduction of the compact disc (CD) marked a turning point in the audio industry toward digital audio storage and processing. Digital technology opens up a great many new possibilities, from reducing the size and increasing the capacity of storage media, to using powerful computer hardware and software to synthesize and manipulate audio in previously unimagined ways. Today, analog audio storage devices are a thing of the past, and analog audio signals are typically employed only where necessary—at the microphone and the speaker.

As we saw in Section 14.2.1.1, the distinction between analog and digital audio technologies corresponds exactly to the distinction between *continuous-time* and *discrete-time* signals in the study of signal processing theory.

14.3.2.1 Analog-to-Digital Conversion: Pulse-Code Modulation

To record audio for use in a digital system, such as a computer or game console, the time-varying voltage of an analog audio signal must first be converted into digital form. Pulse-code modulation (PCM) is the standard method for encoding a sampled analog sound signal so that it can be stored in a computer's memory, transmitted over a digital telephony network or burned onto a compact disc.

In pulse-code modulation, voltage measurements are taken at regular time intervals. The voltage measurements may be stored in floating-point format, or they may be *quantized* so that each measurement can be stored in an integer with a fixed number of bits (typically 8, 16, 24 or 32). The sequence of measured voltage values is then stored into an array in memory, or written out to a long-term storage medium. The process of measuring a single analog voltage and converting it to quantized numeric form is called *analog-to-digital conversion* or A/D conversion. Specialized hardware is typically used to perform A/D conversions. When we repeat this process at regular time intervals, it is called *sampling*. A hardware or software component that performs A/D conversion and/or sampling is referred to as an A/D converter or ADC.

In math terms, given the continuous-time audio signal $p(t)$, we construct the *sampled* version $p[n]$ such that for each sample, $p[n] = p(nT_s)$, where n is a non-negative integer used to index the samples, and T_s is the amount of time between each sample, known as the *sampling period*. The basics of sampling are illustrated in Figure 14.25.

The digital signal that results from PCM sampling has two important properties:

- *Sampling rate*. This is the frequency at which the voltage measurements

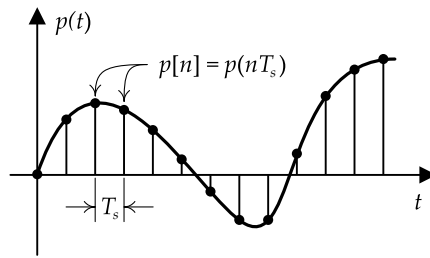


Figure 14.25. A discrete-time signal can be thought of as a sampled version of a continuous-time signal.

(samples) are taken. In principle an analog signal can be recorded digitally without *any* loss of fidelity, provided that it is sampled at a frequency *twice* that of the highest-frequency component present in the original signal. This somewhat astounding and incredibly useful fact is known as the *Shannon-Nyquist sampling theorem*. As we saw in Section 14.1.2.2, humans can only hear sounds within a limited band of frequencies (from 20 Hz to 20 kHz). So all audio signals of interest to human beings are *band-limited*, and can be faithfully recorded using a sampling rate of a little over 40 kHz. (Voice signals occupy a narrower band of frequencies, from 300 Hz to 3.4 kHz, so digital telephony systems can get away with a sampling frequency of only 8 kHz.)

- *Bit depth*. This describes the number of bits used to represent each quantized voltage measurement. *Quantization error* is the error introduced by rounding the measured voltage values to the nearest quantized value. All other things being equal, a greater bit depth results in lower quantization error, and therefore yields a higher-quality audio recording. A bit depth of 16 is typical among uncompressed audio data formats. Bit depth is sometimes referred to as *resolution*.

The Shannon-Nyquist Sampling Theorem

The *Shannon-Nyquist sampling theorem* states that if a *band-limited* continuous-time signal (i.e., a signal whose Fourier transform is zero everywhere outside a limited band of frequencies) is sampled to produce its discrete-time counterpart, the original continuous-time signal can be recovered *exactly* from the discrete signal, provided that the sampling rate is high enough. The minimum sampling frequency for which this relation holds is called the *Nyquist*

frequency.

$$\omega_s > 2\omega_{\max},$$

where

$$\omega_s = \frac{2\pi}{T_s}.$$

Clearly, it is the existence of this theorem that allows digital technology to be used in audio processing. Without it, digital audio would be doomed never to sound as good as analog audio, and computers would not be playing the significant role in the production of high-fidelity audio that they do today.

We won't get into all of the gory details of why the sampling theorem works. But we can gain some insight by realizing that the act of sampling a signal at regularly spaced intervals in time causes its frequency spectrum (Fourier transform) to be duplicated over and over along the frequency axis. The higher the sampling frequency, the more "spaced out" these copies of the signal's frequency spectrum will be. So if the original signal is band-limited,

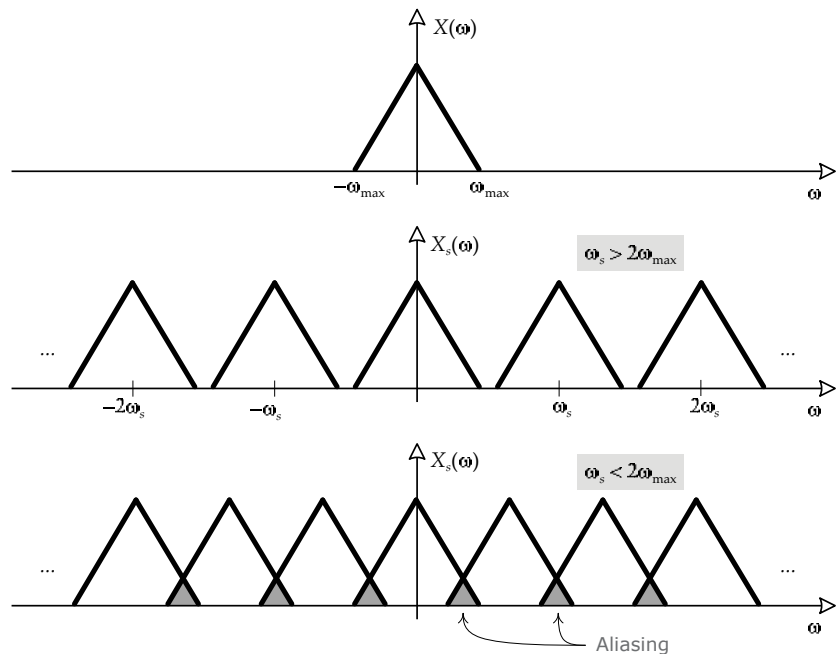


Figure 14.26. The frequency spectrum of a band-limited signal is zero everywhere except within a limited frequency band (top). If the sampling frequency exceeds the Nyquist frequency, the spectrum copies do not overlap and the original signal can be recovered exactly (middle). If the sampling frequency is too low, the spectrum copies overlap and aliasing results (bottom).

and if the sampling frequency is high enough, we can guarantee that the copies of the frequency spectrum will be spaced far enough apart so as not to overlap with one another. When this happens, we can recover the original frequency spectrum exactly via a low-pass filter that filters out all of the copies of the spectrum except the original. However, if the sampling frequency is too low, the spectrum copies will overlap with one another. This is called *aliasing*, and it prevents us from exactly recovering the original signal's spectrum. See Figure 14.26 for an illustration of aliased and unaliased sampling.

14.3.2.2 Digital-to-Analog Conversion: Demodulation

When a digital sound signal is to be played back, a process opposite to that of analog-to-digital conversion is required. We call this, sensibly enough, *digital-to-analog conversion* or D/A conversion for short. It is also termed *demodulation* because it undoes the effects of pulse-code modulation. A digital-to-analog conversion circuit is called a DAC.

D/A conversion hardware generates an analog voltage corresponding to each sampled voltage value in a digital signal, as represented by an array of quantized PCM values in memory. If we drive this hardware with new values periodically, at the rate at which the samples were measured during PCM, and presuming that the sample rate was high enough as per the Shannon-Nyquist sampling theorem, the analog voltage signal produced should exactly match the original voltage signal.

Practically speaking, when we drive an analog voltage circuit with a sequence of discrete voltage levels, unwanted high-frequency oscillations are often introduced as the hardware tries to rapidly change from one voltage level to another. D/A hardware typically includes a low-pass or band-pass filter to remove these unwanted oscillations, thereby ensuring an accurate reproduction of the original analog signal. For more information on filtering, see Section 14.2.5.8.

14.3.2.3 Digital Audio Formats and Codecs

Various data formats exist for storing PCM audio data on disc or transmitting it over the Internet. Each format has its history, and its pros and cons. Some formats such as AVI are actually “container” formats, which can encapsulate digital audio signals in more than one data format.

Some audio formats store the PCM data in an uncompressed form. Others utilize various forms of data compression to reduce the required file size or transmission bandwidth. Some compression schemes are *lossy*, meaning that some of the fidelity of the original signal is lost in the compression/decompression process. Other compression schemes are *lossless*, meaning that

the original PCM data can be recovered exactly after a round-trip compression/decompression cycle.

Let's take a look at a few of the most common audio data formats.

- *Raw header-less PCM* data is sometimes used in situations where the meta-information about the signal, such as the sample rate and bit depth, is known a priori.
- *Linear PCM* (LPCM) is an uncompressed audio format that can support up to eight channels of audio at a 48 kHz or 96 kHz sampling frequency, and 16, 20 or 24 bits per sample. The “linear” in LPCM refers to the fact that the amplitude measurements are taken on a linear scale (as opposed to, say, a logarithmic scale).
- WAV is an uncompressed file format created by Microsoft and IBM. Its use is commonplace on the Windows operating system. Its correct name is “waveform audio file format” although it is also rarely referred to as “audio for windows.” The WAV file format is actually one of a family of formats known as *resource interchange file format* (RIFF). The contents of a RIFF file are arranged in chunks, each with a four-character code (FOURCC) that defines the contents of the chunk and a chunk size field. The bitstream in a WAV file conforms to the linear pulse-code modulation (LPCM) format. WAV files can also contain compressed audio, but they are most commonly used for storing uncompressed audio data.
- WMA (Windows Media Audio) is proprietary audio compression technology designed by Microsoft as an alternative to MP3. See http://en.wikipedia.org/wiki/Windows_Media_Audio for details.
- AIFF (audio interchange file format) is a format developed by Apple Computer, Inc. and used widely on Macintosh computers. Like a WAV/RIFF file, an AIFF file typically contains uncompressed PCM data, and is comprised of chunks, each prefaced by a four-character code and a size field. AIFF-C is a compressed variant of the AIFF format.
- MP3 is a lossy compressed audio file format that has become the de facto standard on most digital audio players, and is also widely used by games and multimedia systems and services. The full name of this format is actually MPEG-1 or MPEG-2 audio layer III. MP3 compression can result in files that are one-tenth the size, but with very little perceptual difference from the original uncompressed audio. These results are achieved by making use of *perceptual coding*—a technique that eliminates portions of the audio signal that are beyond the perception of most people anyway.

- *ATRAC* stands for Adaptive Transform Acoustic Coding—a family of proprietary audio compression techniques developed by Sony. The format was originally developed to allow Sony’s MiniDisc media to contain audio with the same running time as a CD while occupying significantly less space and undergoing an imperceptible degradation in quality. See http://en.wikipedia.org/wiki/Adaptive_Transform_Acoustic_Coding for more details.
- *Ogg Vorbis* is an open source file format that offers lossy compression. Ogg refers to a “container” format that is commonly used in conjunction with the Vorbis data format.
- *Dolby Digital (AC-3)* is a lossy compression format supporting channel formats from mono to 5.1 surround sound.
- *DTS* is a collection of theater audio technologies developed by DTS, Inc. DTS Coherent Acoustics is a digital audio format transportable through S/PDIF interfaces (see Section 14.3.2.5) and used on DVDs and Laserdiscs.
- *VAG* is a proprietary audio file format available for use by all PlayStation 3 developers. It makes use of *adaptive differential PCM* (ADPCM), an analog-to-digital conversion scheme based on PCM. Differential PCM (DPCM) stores the deltas between samples rather than the absolute values of the samples themselves, in order to allow the signal to be compressed more effectively. Adaptive DPCM varies the sample rate dynamically in order to further improve the achievable compression ratio.
- *MPEG-4 SLS, MPEG-4 ALS and MPEG-4 DST* are formats that offer lossless compression.

This list is by no means comprehensive. In fact, there are a dizzying number of audio file formats, and an even longer list of compression/decompression algorithms. For an introduction to the fascinating world of audio data formats, check out our old friend Wikipedia: http://en.wikipedia.org/wiki/Digital_audio_format. The “PlayStation 3 Secrets” website also provides some excellent information on audio formats: <https://bit.ly/2HOVtvR>.

14.3.2.4 Parallel and Interleaved Audio Data

One way to organize multi-channel audio data is to store the samples for each monophonic channel into a separate buffer. In this case, you’d need six parallel buffers to describe a 5.1 audio signal. This arrangement is shown in Figure 14.27.

Multi-channel audio data can also be *interleaved* within a single buffer. In this case, all of the samples for each time index are grouped together in a pre-

Interleaved

C[n]
FL[n]
FR[n]
RL[n]
RR[n]
LFE[n]
C[n+1]
FL[n+1]
FR[n+1]

...

Figure 14.28. Six-channel (5.1) PCM bus data in interleaved format.

defined order. Figure 14.28 depicts an interleaved PCM buffer containing a six-channel (5.1) audio signal.

14.3.2.5 Digital Wiring and Connectors

S/PDIF (Sony/Philips Digital Interconnect Format) is an interconnect technology that transmits audio signals *digitally*, thereby eliminating the possibility of noise being introduced by analog wiring. The S/PDIF standard is physically realized either via a coaxial cable connection (also called S/PDIF) or a fiber-optic connection (known as TOSLINK).

Regardless of the physical interface (S/PDIF coaxial or TOSLINK optical), the S/PDIF transport protocol is limited to 2-channel 24-bit LPCM uncompressed audio at standard sampling rates ranging from 32 kHz to 192 kHz. However, not all equipment works at all sample rates. The same physical interfaces can be also be used to transport bitstream-encoded audio (e.g., Dolby Digital or DTS lossy compressed data) at bitrates ranging from 32 kbps to 640 kbps for Dolby Digital and 768 kbps to 1536 kbps for DTS, respectively.

Uncompressed multi-channel LPCM (i.e., greater than two stereo channels) can only be sent over an HDMI (high-definition multimedia interface) connection on consumer audio equipment. HDMI connectors are used for transmission of both uncompressed digital video and either compressed or uncompressed digital audio signals. HDMI supports up to a 36.86 Mbps bitrate for multi-channel or bitstream audio. However, HDMI bitrates for audio vary depending on the video mode—only 720 p/50 Hz modes or higher are capable of utilizing the full audio bandwidth. See the HDMI specification under the section “video dependency” for more information on this. Apple’s DisplayPort and Thunderbolt connectors are other high-bandwidth alternatives similar in many respects to HDMI.

USB connections are sometimes used to send audio signals. On most game consoles, the USB output is intended only to drive headphones.

Wireless audio connections are also possible. The Bluetooth standard is the

Parallel

C[n]	FL[n]	FR[n]	RL[n]	RR[n]	LFE[n]
C[n+1]	FL[n+1]	FR[n+1]	RL[n+1]	RR[n+1]	LFE[n+1]
C[n+2]	FL[n+2]	FR[n+2]	RL[n+2]	RR[n+2]	LFE[n+2]
...	...	...	...	...	...

Figure 14.27. Six-channel (5.1) PCM bus data in parallel format.